VIDYAPEETH

BATCH CODE: 12-ANI81MA

SUBJECT NAME: CHEMISTRY

CHAPTER NAME: Structure of atom

Lecture No. 14

By Amit Sir

Today's Goal

Subtopic .. Stopping Potential

Subtopic .. Bohr's Model

Subtopic ..

* effect of Intensity

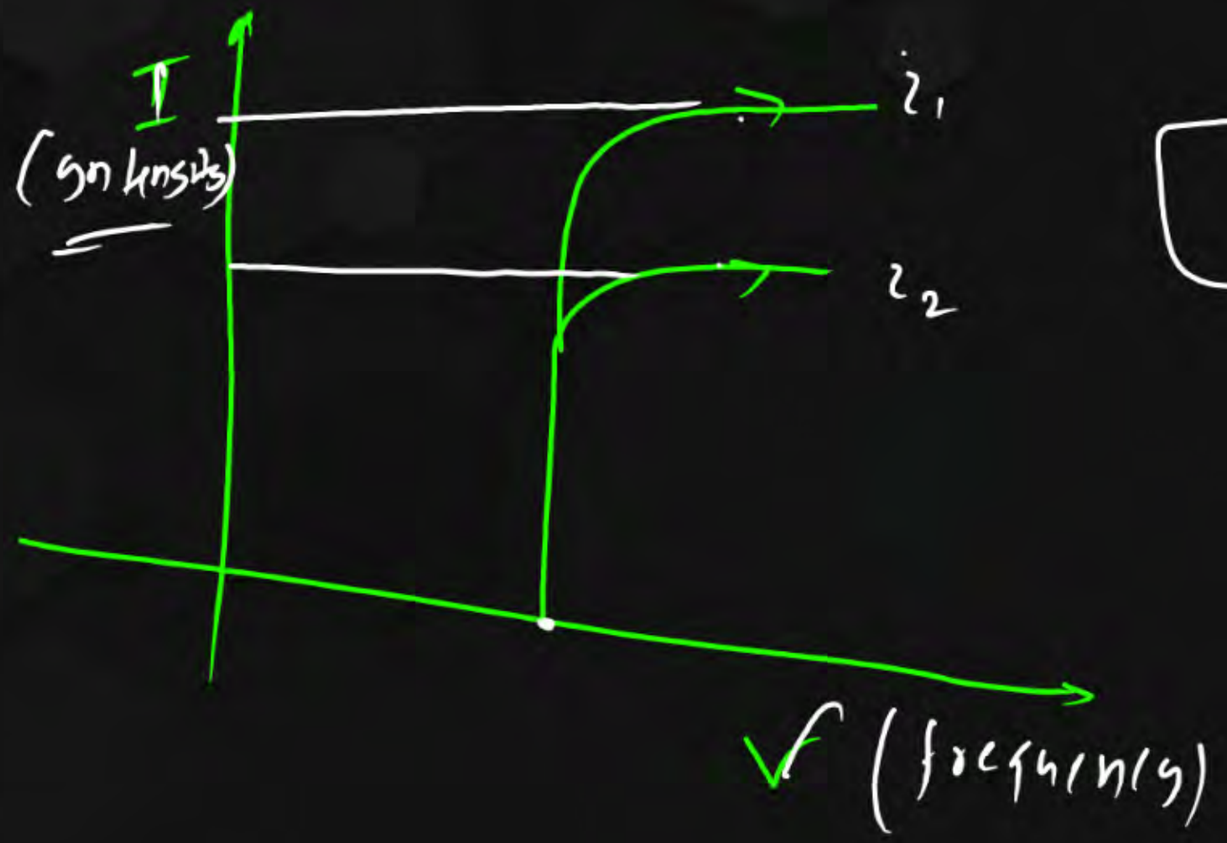
→ no of photons falling per unit area per unit time

$$\boxed{I = \frac{n}{A \cdot t}}$$

n → no. of photons

* Intensity ↑ ⇒ no. of photons ↑ ⇒ no. of $\bar{e}$ ↑ ⇒ Photoelectric current ↑
* Intensity ↓ ⇒ no of photons ↓ ⇒ no. of $\bar{e}$ ↓ ⇒ " " " ↓

Qn



$$\boxed{i_1 > i_2}$$

Effect of Frequency

$\uparrow E = \phi_0 + KE \uparrow$

ϕ_0 → fixed

As Frequency ↑ ⇒ Energy of incident photon ↑

⇒ K.E of photo electron ↑

$\uparrow \boxed{v \propto KE}$

Qn A metal is irradiated by light of frequency 3.2×10^{15} Hz. the photoelectron emitted has its K.E 2 times of the photoelectrons emitted when the same metal was irradiated by 2×10^{15} Hz frequency of light. Find the work function and threshold frequency.

$\phi_0 = h\nu_0$

$K(2\nu_2-\nu_1) = K\nu_0$

$\nu_0 = 2\nu_2-\nu_1$

$KE_1 = 2KE_2$

$\Rightarrow h\nu_1-\phi_0 = 2(h\nu_2-\phi_0)$

$\Rightarrow h\nu_1-\phi_0 = 2h\nu_2-2\phi_0$

$\Rightarrow \phi_0 = 2h\nu_2-h\nu_1$

$= h(2\nu_2-\nu_1)$

$h\nu_1 = \phi_0 + KE_1$

$KE_1 = h\nu_1 - \phi_0$

$h\nu_2 = \phi_0 + KE_2$

$KE_2 = h\nu_2 - \phi_0$

Stopping Potential :→

V_s

→ The minimum value of anode potential which stops the fastest moving electron is called stopping potential.

$W = eV_s$

$h\nu = \phi_0 + KE$

$$h\nu = \phi_0 + eV_s$$

V_s = stopping potential

reex (Qn) when a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. calculate the threshold wavelength for the metallic surface.

(a) 3λ

(b) 4λ

(c) 5λ

(d) $\frac{5\lambda}{2}$

$$\frac{hc}{\lambda} = \phi_0 + eV$$

$$\frac{hc}{2\lambda} = \phi_0 + e\frac{V}{4}$$

$$\frac{2hc}{\lambda} = 4\phi_0 + eV$$

$$\frac{hc}{\lambda} - \phi_0 = \frac{2hc}{\lambda} - 4\phi_0$$

$$3\phi_0 = \frac{hc}{\lambda}$$

$$\Rightarrow 3\frac{hc}{\lambda_0} = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda_0 = 3\lambda$$

#

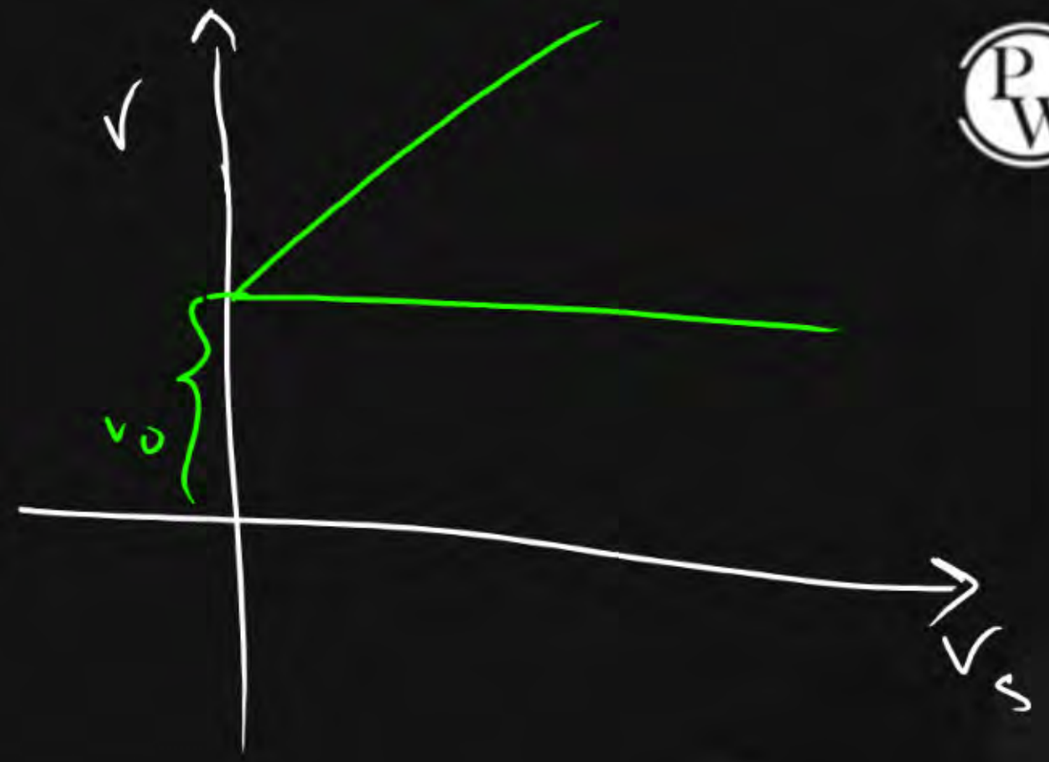
ν vs V_s

y — x

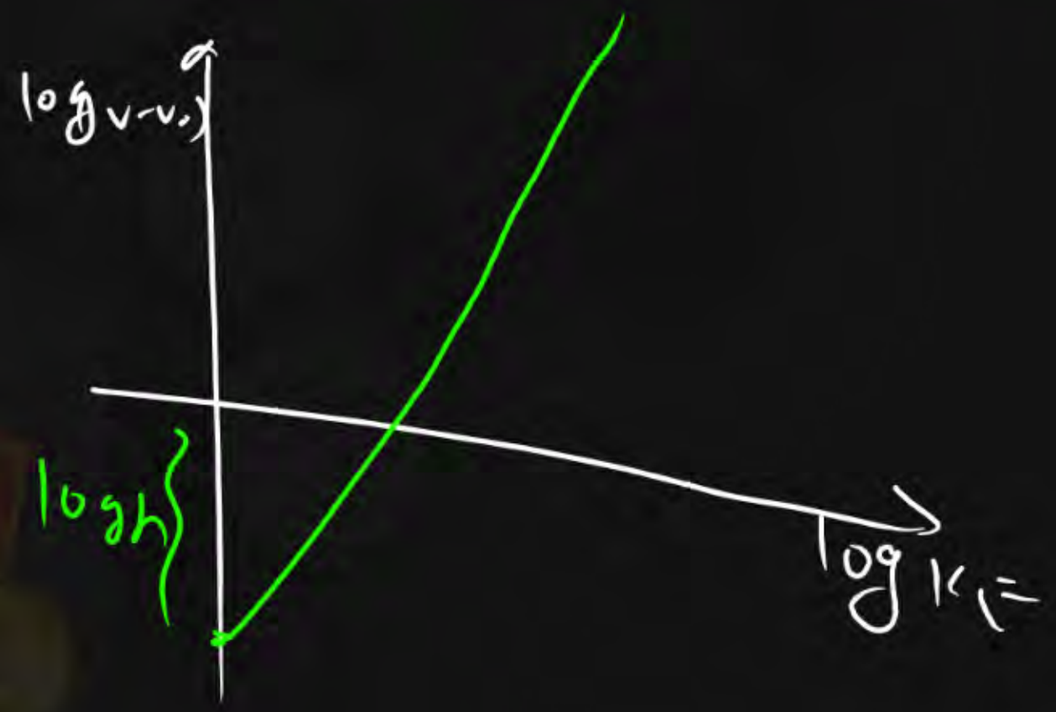
$$h\nu = h\nu_0 + eV_s$$

$$V = V_0 + \frac{e}{h} V_s$$

y — c — x — m



$\log(\nu - \nu_0)$ vs $\log KE$



PPP-4
4, 7, 8, 9, 10

P-29
18, 19, 20

$h\nu = h\nu_0 + KE$

$h(\nu - \nu_0) = KE$

$\nu - \nu_0 = \frac{KE}{h}$

$\log(\nu - \nu_0) = \log\left(\frac{KE}{h}\right)$

$\log(\nu - \nu_0) = 1 \log KE - \log h$

y m x c

$\log(A \times B) = \log A + \log B$

$\log\left(\frac{A}{B}\right) = \log A - \log$

$E = h\nu$

$\log E = \log h + \log \nu$

E vs ν

log E vs log ν

$$\frac{hc}{\lambda} - eV = \frac{hc}{2\lambda} - \frac{eV}{4}$$

$$\Rightarrow \frac{hc}{\lambda} - \frac{hc}{2\lambda} = eV - \frac{eV}{4}$$

$$\Rightarrow \frac{hc}{2\lambda} = \frac{3eV}{4}$$

$$\Rightarrow \lambda = \frac{2hc}{3eV}$$

$$\phi_0 = \frac{hc}{\lambda} - eV$$

$$= \frac{hc}{2hc} \times 3eV - eV$$

$$= \frac{3}{2}eV - eV$$

$$\frac{hc}{\lambda_0} = \frac{1}{2}eV$$

★★★★★

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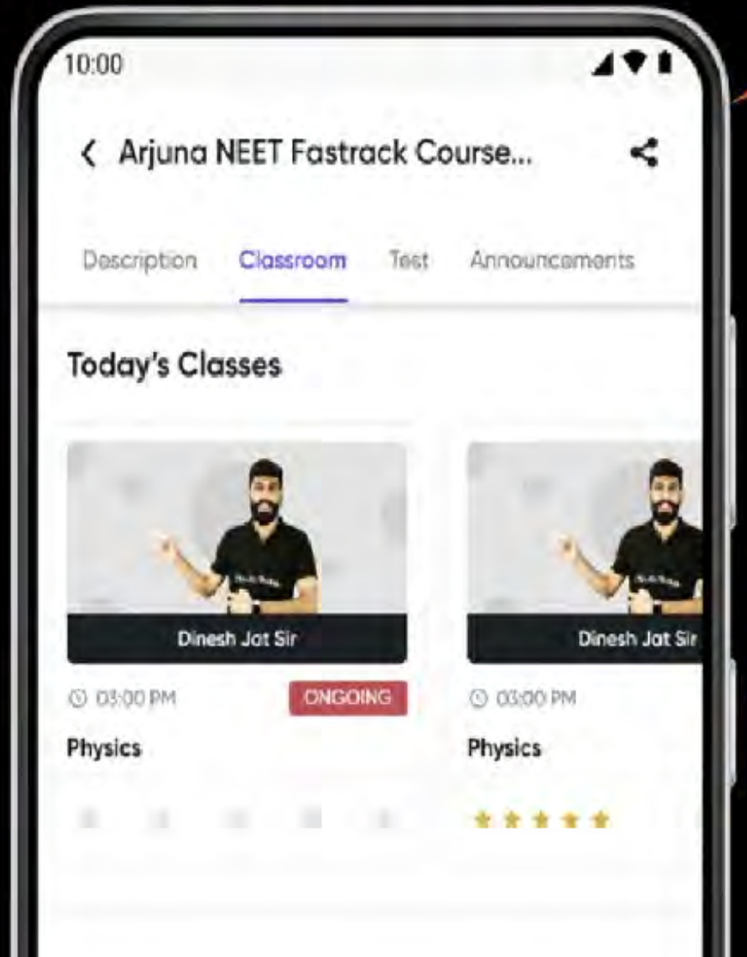
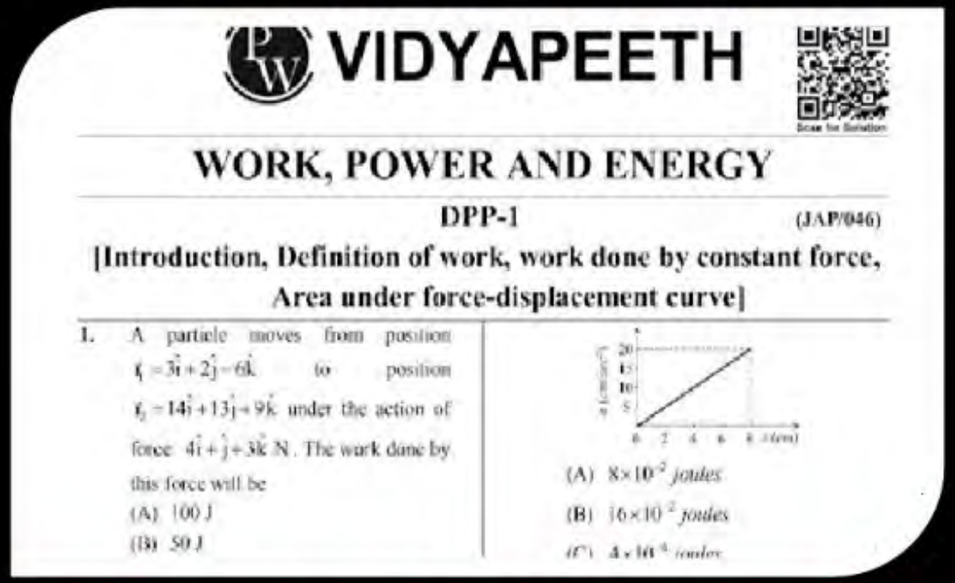


"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



VIDYAPEETH

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} joules

(B) 16×10^{-2} joules

Thank You!!!!

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